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ABSTRACT

We present an approach to analyzing the motion capture data of rowers using bivariate functional principal component analysis (bfPCA). The method has been applied on data from six elite rowers rowing on an ergometer. The analyses of the upper and lower body coordination during the rowing cycle revealed significant differences between the rowers, even though the data was normalized to account for differences in body dimensions. We make an argument for the use of bfPCA and other functional data analysis methods for the quantitative evaluation and description of technique in sports.

CCS CONCEPTS

• **Computing methodologies** → **Motion capture**; • **Mathematics of computing** → *Exploratory data analysis*.

KEYWORDS

Rowing, Motion Capture, Movement Visualization, Functional Data Analysis

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1 INTRODUCTION

Analyzing sports technique is a difficult task, given the multidimensional nature of the human movement, as well as individual differences between athletes (such as body dimensions and proportions). High-quality motion capture recordings can be used to shed light on the multidimensionality of sports technique [9, 12]. Analyzing such data often requires drastic dimensionality reduction and “black box” methods. While such analyses can be effective in, for example, classifying athletes into categories, they may not necessarily give any insight into why and how the technique is shaped the way it is. To answer such questions, functional principal component analysis (fPCA) can be a good method for providing intuitively understandable quantitative metrics.

In this paper we investigate how fPCA can be applied to motion capture data of rowing technique. We are particularly interested in understanding more about:

- What is the dimensionality of the rowing technique in our sample?
- How does the technique differ across rowers?
- What are the benefits of using fPCA over dimensionality reduction methods such as PCA?

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Figure 1: One of the athletes using the rowing machine in the motion capture lab. Reflective markers can be seen on the main joints of the body.

We will start by giving an overview to rowing and the analysis of rowing technique before presenting our data set consisting of motion capture data of six elite rowers. Then, we will analyze this data to illustrate the insights that can be achieved with fPCA.

2 BACKGROUND AND RELATED WORKS

Rowing can be defined as a cyclical motion, where the rower alternates between two phases: (i) a “drive” phase, during which the rower is applying force to propel the boat forward, and (ii) a “recovery” phase, during which the rower is returning to the start position without applying propelling force. The first phase starts at the “catch” position, where the rower is “crouched”, with knees bent and torso leaning forward. From there, the rower applies force to the oars through the sequential straightening of knees and back, and finally by pulling the arms close to the chest. This completes the first part of the cycle, and the rower is now at the “finish” position, and cannot apply more propulsive force. The rower then starts the recovery phase, by lifting the oar blades out of the water and going through approximately the same body movement in reverse, ending in the crouched catch position, in order to begin a new cycle [5]. We use the term “approximately” because there are some differences in the coordination during drive and recovery, which can be seen in the left part of Figure 2. One such difference is the “rock over” movement during the recovery phase, where the athlete starts swinging the back forward before they start to bend their knees. This is in contrast to the drive phase, where both the knee and back angles increase together.

The specific coordination and magnitude of these body movements is what constitutes an athlete’s rowing technique, and improving a rower’s performance by tweaking their technique is the subject of several studies in sports biomechanics. Such studies often analyze the relation between propulsive force and oar angle

[13, 15–18]. In some works that investigate the changes in movement under different experimental conditions, the focus is on mean velocity or intra-cycle velocity fluctuations as comparison metrics [3, 4, 10, 11]. All of these performance metrics are the final results of, among other factors, the athlete’s movement (technique), but are not measurements of the movement itself. To investigate the movement pattern directly, the athlete’s whole body has to be recorded and analyzed. Full-body motion capture can be used for such biomechanical investigations, although it generally requires that the subjects row on a rowing machine—in a lab setting—since performing full-body motion capture on an athlete sitting in a boat on water is very challenging. Rowing on a machine is biomechanically similar to on-water rowing, with most differences occurring at the upper-body motion, especially at the arms [14]. Nonetheless, it enables a closer view of the athlete’s movement than the boat instrumentation used in the aforementioned works.

In the following sections, we will explore previous research on rowing technique and elaborate on the methods and findings of studies of rowing and human motion.

2.1 Rowing technique

In a comprehensive review of the research on rowing technique, Soper and Hume [14] note that, although technique coaching is often emphasized in training programs, there are no clear guidelines as to how this technique should look like for individuals of different body dimensions.

There is widespread agreement that force-time profiles similar to those of elite rowers are indicators of enhanced performance. A force-time profile the curve of propulsive force in relation to the duration of a single rowing cycle [13]. There is, however, limited research on how to acquire such force-time profiles in the first place, i.e. how to tweak technique for better performance. It is believed that more important than improving muscle strength, is the improvement of rowing technique as the effective sequencing of lower-body, upper-body, and arms movement [14]. Several approaches to this sequencing have been proposed, such as legs-first (“Rosenberg style”) or trunk-first.

Movement sequencing can be described by the relationship between the knee angle and the back angle during the rowing cycle, as can be seen in Figure 2. Angle-angle diagrams are commonly used as a measure of body segment coordination, especially due to their time-invariance (within and between subjects) and intuitive interpretation [6]. For our purpose, angle-angle functions allow for statistically comparing movement patterns using functional data analysis.

2.2 Functional data analysis in rowing

Functional data analysis (FDA) is a set of statistical methods for analyzing correlated data, for example, the temperature recorded at a weather station as a *function* of time. In this paper we focus on the application of one such method, called *functional* principal component analysis (fPCA). Much like the non-functional version of principal component analysis (PCA), fPCA is employed for exploring data sets. By leveraging the functional nature of the data, it can provide advantages over its non-functional counterparts. Regular PCA can be used to find a way to represent data that highlights the

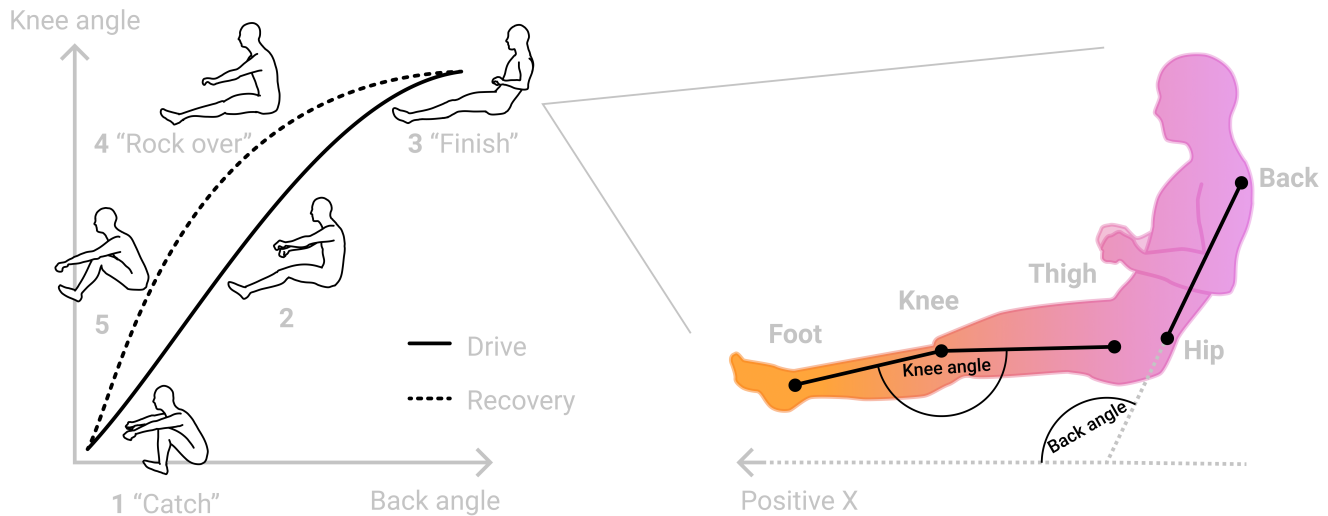


Figure 2: Left: a curve representing the rowing cycle in terms of knee and back angles. The solid line represents the drive phase, and the dashed line the recovery phase. The rower position in several points along the curve is illustrated. Right: A rower in the “finish position” of the rowing cycle highlighting the two angles analyzed as well as the position of the markers used to calculate them. Illustrations based on the guidelines of the World Rowing Federation [5], curve from our own dataset.

most amount of variation between each data point, that is, it shows the principal *directions* of variation. Functional PCA, in addition to that, helps finding the fundamental *modes* of variation. To illustrate this difference, the PCA of all the temperature measurements of several weather stations during one year would allow for finding out which weather stations are more alike (have closer values for their principal components) and which are less alike (have farther values for their principal components). This can be done by looking at just a few values instead of comparing every daily temperature measurement as you would do with other statistical methods. Functional PCA goes further by also providing the function that explains this variation. For example, two weather stations could be explained to be closer to each other because they both score high on the first principal mode of variation, which happens to describe a higher variation in temperature during the summer and a smaller one during the winter [7].

An intuitive way to understand functional principal components is given by Amelynck et al. [1]: they represent a low dimensional model that, combined with the average of all curves, can be used to describe each single curve through a simple linear combination. Therefore, the functional principal component (fPC) scores of a curve represent how close it is to the respective principal component functions.

FDA enjoys growing popularity in the field of biomechanics. One reason for this may be that it is hard to analyze the multidimensional and continuous time series of human movement data with conventional statistical methods. FDA allows for modelling such spatiotemporal data as a set of functions. Some studies have analyzed the rowing technique using the fPCA technique, as well as its bivariate version (bfPCA). Warmehoven et al. [18] analyzes the propulsive force and oar angle of rowers on-water using both fPCA on each variable separately, and bfPCA for a joint analysis.

Stepwise discriminant function analysis yielded good results for the classification of female rowers and domestic level athletes: 90.7% (fPCA), 92.6% (bfPCA) and 87.5% (fPCA), 85% (bfPCA), respectively. Results for the classification of higher-skilled and male rowers were worse. The authors believe that the principal components could be used to inform discussion about technique between athletes and coaches.

In a later study on only the female rowers in their previous dataset, Warmehoven et al. [17] present the first 5 bivariate functional principal components (bfPC's) for bow and stroke side (right and left side of the rowers, respectively) force-angle curves, which account for 95% of curve variation in both cases. Bivariate correlation tests were carried out between the bfPC's and boat speed. This was followed by uni-variate ANOVA to test for significant differences between mean bfPC scores in different competition levels. Finally, a linear discriminator was used to classify each force-angle curve according to competition level, with an accuracy of 74.1% and 59.3% for bow and stroke side respectively. The authors also discuss several strategies for data presentation of the rowing cycle for fPCA and bfPCA in [16] and [15].

2.3 Functional data analysis of human movement data

Apart from rowing, there are also several studies using functional data analysis on human movement. Harrison et al. [6] employed bfPCA to calculate biomechanically interpretable principal component functions of vertical jumping using knee-hip, knee-ankle, and hip-ankle angle pairs. A linear discriminator model is also presented, with a classification accuracy of 67.3% (knee-hip), 63.3% (knee-ankle), and 69.4% (hip-ankle) into three developmental stages.

In research on music-related movement, Amelynck et al. [1] used fPCA to analyze hand and chest speed of subjects listening to a piece of music. They found that 70% of the movement variability was retained with 1 (chest) and 3 (hand) fPC's. This low number of principal components explaining most of the variability was interpreted as the music being perceived and embodied in similar ways among subjects. Interestingly, the fPC scores are regarded as the “individualities” in the motion of each subject in relation to the “commonality” of the average motion of all of them. Moreover, fPCA is seen as useful because regular multivariate principal component analysis would have had to be applied on 315,000 (315×100 Hz) data points per marker—when using only marker speed—which is computationally expensive. Assuming that the data is related over samples, FDA can be employed on the dataset decomposed into a few basis functions, which is simpler and also renders a more intuitively understandable result.

3 PUTTING IT ALL TOGETHER: INVESTIGATING ROWING TECHNIQUE WITH FDA

Our current study is similar to the research of Amelynck et al. [1] in that it investigates body motion under different conditions directly with FDA. While they studied dance, our focus is on rowers, which is a much more restricted “motion manifold”. This means that we can analyze position data almost directly, while they had to rely on velocity profiles since positional data varies greatly in free expressive movement. We also build on Warmehoven's et al. extensive research of rowing using functional data analysis [15–18] by using high-dimensional motion capture data of full-body movement instead of boat instrumentation. Finally, we are inspired by Harrison's et al. work [6] also because of this capture method, instead of the laborious digitization of a single view 2D camera recording. Moreover, we believe that our analysis could help inform rowing practice design and act as a concrete measurement of how the technique of a rower changes over time, or how it compares to that of another rower.

In this section, we describe the data set used in our analysis. Then, we analyze the data set and illustrate how the motion capture data can be made sense of using FDA.

3.1 Rowing motion capture dataset

Our analysis was conducted on a data set of rowing that contained full-body motion capture of athletes performing on a rowing machine. Data from 9 subjects was captured, but data of 3 subjects was not used due to the occlusion of certain markers on the recordings or due to the wrong detection of markers on reflective clothing. This left us with a data set of 6 rowers ($n=6$, all-male, mean age= 23.17 ± 3.87 years, mean height= 1.83 ± 0.06 m). The athletes were asked to row, one at a time, on a Concept 2 rowing machine at a stroke rate of 26 strokes per minute. The drag factor of the rowing machine could be freely adjusted by each athlete. The movement of the subjects was recorded using a Qualisys motion capture system with 9 Oqus 300 cameras. Passive markers were attached to the subjects' hands, elbows, shoulders, back, hips, thighs, knees, and feet, as well as on the handles of the ergometer. The setup can be seen in Figure 1. The sampling rate of the recordings was 200 Hz.

The athletes rowed for 3 minutes, and 40 consecutive rowing cycles were extracted from each of the rowers after the athlete had reached a steady pace (SD of cycle length inferior to 50 ms). The rowing cycles were defined using a peak detection algorithm on the trajectory of the back marker on the longitudinal axis of the movement. In total, 240 cycles were analyzed.

To analyze movement coordination between the upper and lower body of the rowers, the back and knee angles were extracted from the recordings using the Motion Capture Toolbox [2]. The knee angle was extracted from both knees as the angle between the foot-knee and the knee-thigh segments. Then the average of the right and left knee angles at each time step was used for the analysis. The back angle was calculated as the angle between the vector going through the back marker and the midpoint of the subject's hips and an infinite vector pointing towards the front of the rowing machine and parallel to its longitudinal axis (Figure 2, right). Using these two angles greatly reduces the dimensionality of the dataset, but it still leaves 72,000 data points per subject ($2 \text{ angles} \times 180 \text{ s} \times 200 \text{ Hz}$). To mitigate individual differences between the subjects, the angles were each amplitude normalized to a range of 0 to 100, where 100 is the cycle's maximum angle. For both the knee and the back angles, this corresponds approximately to the “finish” position on the end of the drive phase, when the rower is “laying back” (as shown in Fig 2, right). In order to perform the functional data analysis, the cycles were also length normalized (in terms of time), which seemed appropriate given that all athletes were performing at the same stroke rate [16].

The processed data was transformed into functions and analysed using Ramsay's fda package for MATLAB [8]. First, the discrete measurements were approximated to smooth curves for each rowing cycle. This resulted in 240 curves for each of the angles, which can be plotted as an angle-angle diagram, such as in Figure 2. Then, bivariate functional principal component analysis was performed on the curves similar to Harrison et al. [6] and Warmehoven et al. [15].

3.2 Results

The 4 first bfPC's accounted for 96.3% of the set's variability, and can be seen, together with the mean of all cycles, in Figure 3. The bfPC scores of each cycle were used as dependent variables on a multiway non-parametric comparison with the subject as the grouping variable. First, the non-normality of the data was assessed with a Shapiro-Wilk test. Then, performing a Kruskal-Wallis test, the results were found significant for all mean harmonic scores, indicating that the subject's technique did vary significantly across athletes. To pinpoint exactly which pairs of subjects were significantly different, a post-hoc Dunn's test was performed.

Our results (summarized in Table 1) show that the mean bfPC scores were significantly different between most subject pairs. This is despite the fact that the data was normalized to a certain extent, to mitigate differences in individual body dimensions that could further discriminate among rowers. This is in line with the theory of a movement “signature” in elite athletes [16]. Significance levels for the mean scores on bfPC1: <0.05 except for the pairs 1-4 and 4-6. bfPC2: <0.01 except for 1-3, 4-5, 4-6 and 5-6. bfPC3: <0.001 except for 1-4, 1-5, 2-3 and 4-5. bfPC4: <0.01 except for 1-5 and 2-3.

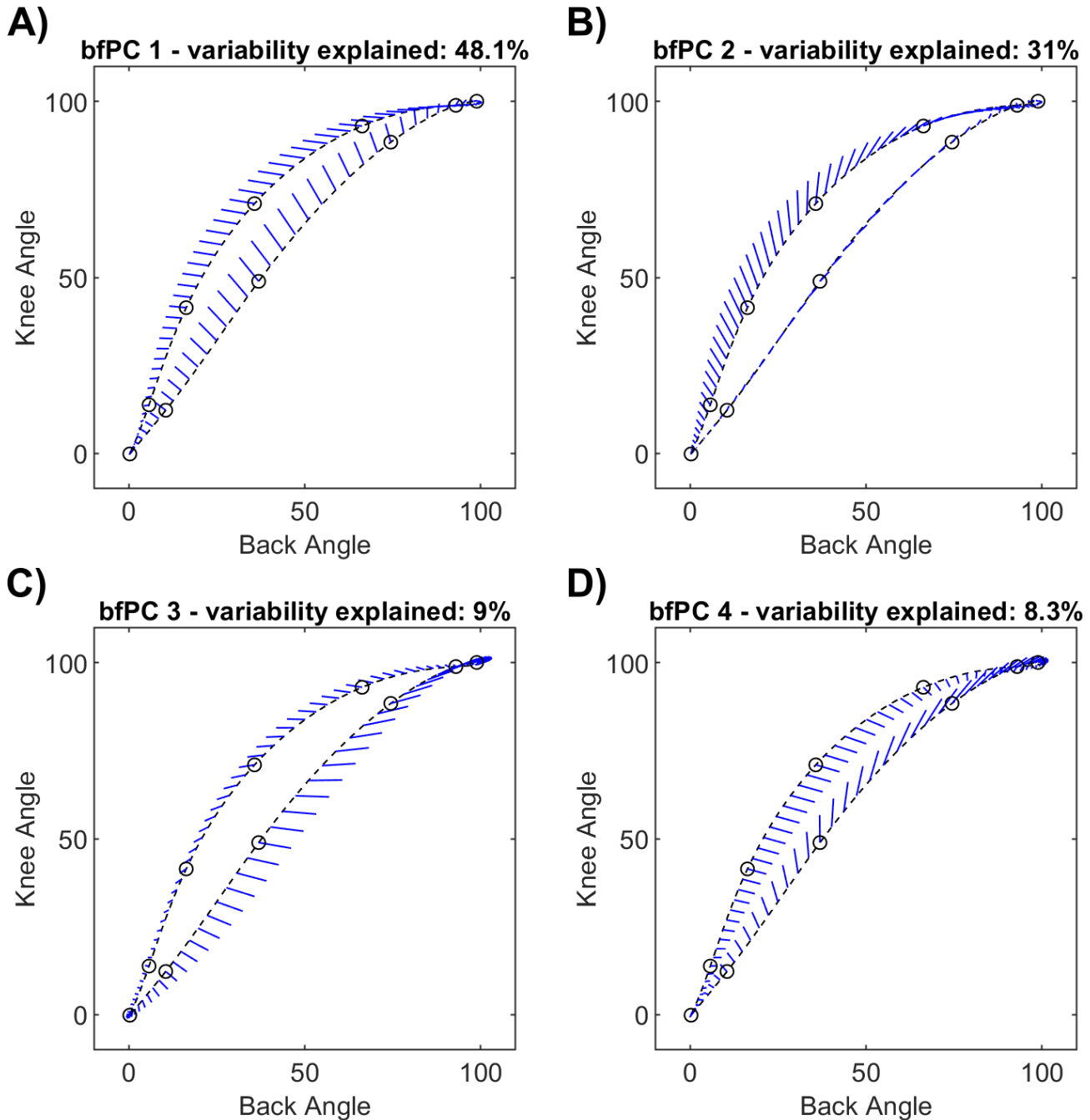


Figure 3: The first four bivariate functional principal components of the 240 rowing cycles analyzed. The dashed line represents the mean cycle, and the solid blue lines pointing from it represent where the bfPC curve would be, multiplied by a suitable constant for visibility. A circle is plotted on the mean curve at each tenth of the cycle

Plotting the first and second bfPC scores of all cycles in a scatter plot shows how the variability between athletes forms well-defined clusters (Figure 4). These conclusions could also have been drawn from applying regular PCA. For example, from just the scatter plot we can see that subjects 1 and 5 are “complete opposites” in

terms of the first two PC scores. The advantage of bfPCA is that it permits understanding what differentiates the subjects in terms of their movement. Such analysis can be carried on by inspecting the functional principal components themselves, and how they affect the mean curve. Therefore, bfPCA can help us understand what

	bfPC1	bfPC2	bfPC3	bfPC4
Subject 1	-4.51 (0.94)	-4.80 (1.02)	-0.66 (0.81)	-0.51 (0.82)
Subject 2	2.19 (0.81)	0.26 (0.87)	0.32 (0.79)	0.19 (1.06)
Subject 3	3.85 (0.63)	-3.65 (0.68)	0.50 (0.77)	0.46 (0.67)
Subject 4	-3.71 (0.75)	2.84 (0.84)	-0.79 (0.63)	2.56 (0.49)
Subject 5	5.36 (1.48)	2.43 (0.90)	-1.36 (2.95)	-0.69 (1.62)
Subject 6	-3.18 (0.67)	2.92 (0.53)	1.99 (0.48)	-2.01 (0.42)

Table 1: Mean (SD) bfPC scores for each subject.

the difference in technique actually is, and, if deemed necessary, inform decisions towards changing it.

We illustrate this in the next section by analyzing why some pairs of subjects are close and others are apart in their bfPC scores. The analysis tasks that we tackle are:

- Understanding what makes one individual deviate from the “mean” technique of the data set.
- Analyzing the technique of two individuals with opposite scores.
- Analyzing the technique of two individuals with similar scores.

3.3 Analyzing technique differences with bfPCA

Let us first examine the mean curve of Subject 5, and how this individual’s technique differs from the “mean technique” of the sample (Figure 5). Having a high positive score on the first bfPC (Figure 3), this subject deviates from the mean both on the drive phase—with it being shifted towards a non-linear knee to back opening coordination—as well as on the recovery phase, with the horizontal plateau on the beginning of the recovery being more pronounced than average. This indicates that the subject keeps the knees extended longer while the back starts to return. This individual is also a positive scorer for the second bfPC, which further contributes to the round shape of the recovery phase. The negative score on bfPC 3 has the opposite effect of that shown in Figure 3, what makes the drive phase deviate even more from a linear relationship between back and knees. The low score on the fourth bfPC is harder to relate to a specific feature of the curve.

We can also compare Subject 5 to Subject 1, who has drastically different scores on the first two bfPC’s. The “individualities” of the two athletes can be analyzed by plotting their mean curves together, as well as the mean curve of the whole set (Figure 6). Subject 1 is a high negative scorer on the first bfPC, what translates into the drive phase being shifted towards an almost 1:1 knee to back opening ratio (the drive phase is nearly a 45 degrees line on the plot) and a softer “bump” on the recovery phase. As mentioned above, Subject 5 is a high positive scorer on the first bfPC, which makes his coordination pattern much less linear.

Finally, we can analyze subjects that are very close in terms of bfPC scores, and see how their techniques relate to one another. In the scatter plot in Figure 4, Subjects 4 and 6 are close in terms of the first two bfPC’s. By plotting their mean curves together, as well as the sample’s mean curve, we can see how the remaining bfPC’s set them apart (Figure 7). Subject 4 is a negative scorer on the

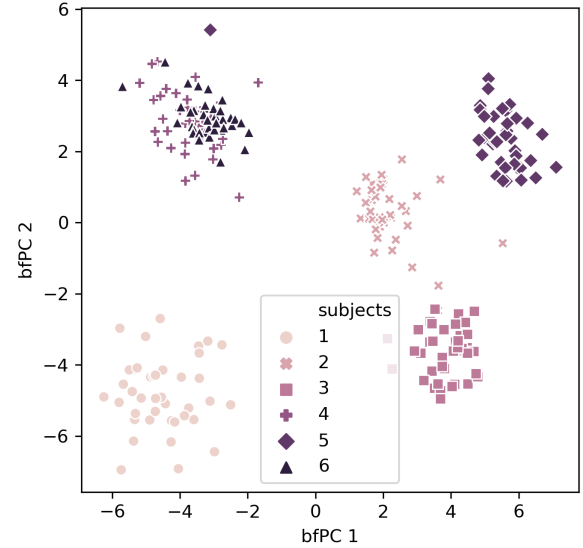


Figure 4: Scatterplot of all cycles in terms of their first and second bfPC scores, labeled by subject.

third harmonic, while Subject 6 is a positive scorer. This explains why the former has a “flatter” drive phase, and the latter has a more curved one, expanding outwards. This is further helped by the high score of Subject 4 on the fourth bfPC, which makes the whole rowing pattern more tightly packed. The opposite is true of Subject 6, who scores negatively on that harmonic, and thus has a looser coordination pattern around the mean and a more pronounced inflection on the drive phase.

This illustrates how an athlete’s technique in terms of back and knee coordination can be analyzed in a drastically reduced, yet still meaningful, 4-dimensional space. These tasks can also be accomplished to some extent by looking up the respective bfPC scores, shown in Table 1. In our sample, the variability in the movement between-athletes was greater than the variability within the same athlete (as is explicit in Figure 4). In other words, the bfPCA mostly captured the individuality of each subject — which is what explains the most variation between all cycles — but it also retained some intra-individual variability. This is also supported by the statistical analysis presented earlier in this section. Analysis of intra-individual variability can also be useful to understand where the technique variability is stemming from, and whether it is desirable or not.

4 FINAL COMMENTS

We have applied bivariate functional PCA to assess differences in rowing patterns for six elite rowers. Significant differences were found across subjects for all bfPC’s. Some of the bfPC’s could be directly related to a movement feature called rock over, which is a common landmark of proper rowing technique. However, it is not clear how the differences in technique identified through bfPCA

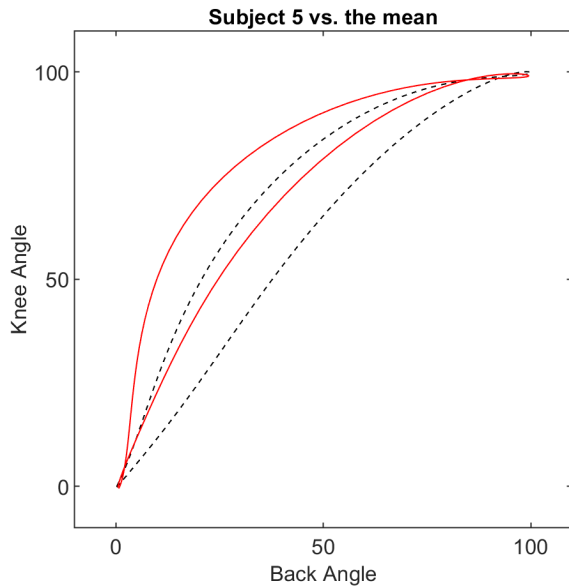


Figure 5: Mean curve for Subject 5 (solid red) and overall mean of all subjects (dashed black).

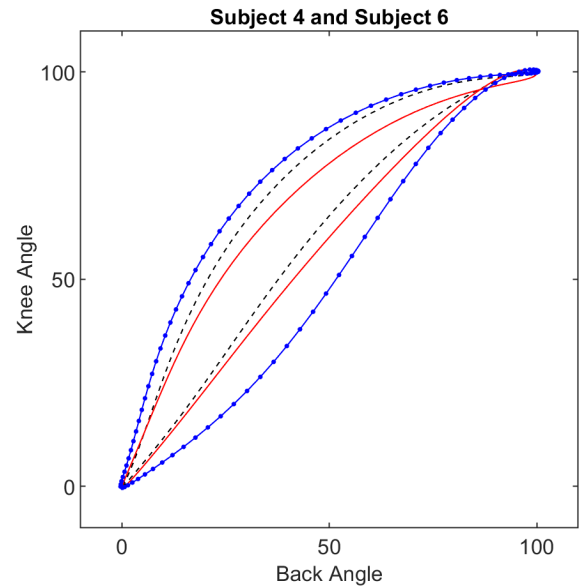


Figure 7: Comparison of Subjects 4 (solid red) and 6 (solid dotted blue) mean curves against the overall mean (dashed black).

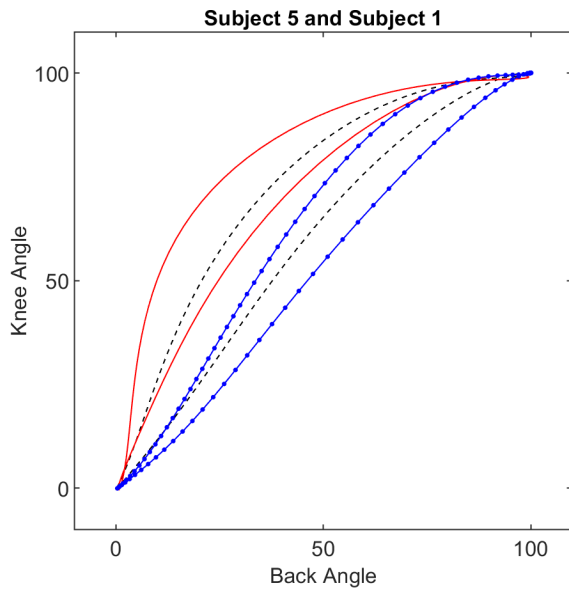


Figure 6: Comparison of Subjects 5 (solid red) and 1 (solid dotted blue) mean curves against the overall mean (dashed black).

among athletes relate to power output or overall performance. To uncover this, a more comprehensive study, ideally on-water, would be necessary. Doing such a study would close the loop from subtle motion analysis to overarching performance metric.

We believe that functional data analysis can be a useful tool for the evaluation of sports practice design, acting as a robust metric while still retaining easy biomechanical interpretability. Specifically, we found that the rowers' movement in our sample could be explained in a low-dimensional model, and that individual differences in technique across rowers explained most of the variation on the data.

The four-dimensional space identified in this work is particular to this fairly homogeneous group of rowers. Adding more rowers of varying age, gender, and skill level would permit verifying if the same low-dimensional space still holds for a larger sample. One possible use of the analysis techniques presented here is on the evaluation of experiments of different practice settings, and how they affect the rowing technique.

Finally, more research is needed to gauge how easy it is to change the parameters (in terms of bfPC scores) of one's technique, and how they are intertwined to one another. In the same vein, a task-oriented evaluation with rowers and coaches could attest to how this technique could inform rowing technique practice outside of the lab.

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